MID-TERM EXAMINATION

(Course Name : B.Tech CSE/ECE) (Semester II)

(May, 2023)OFF LINE mode

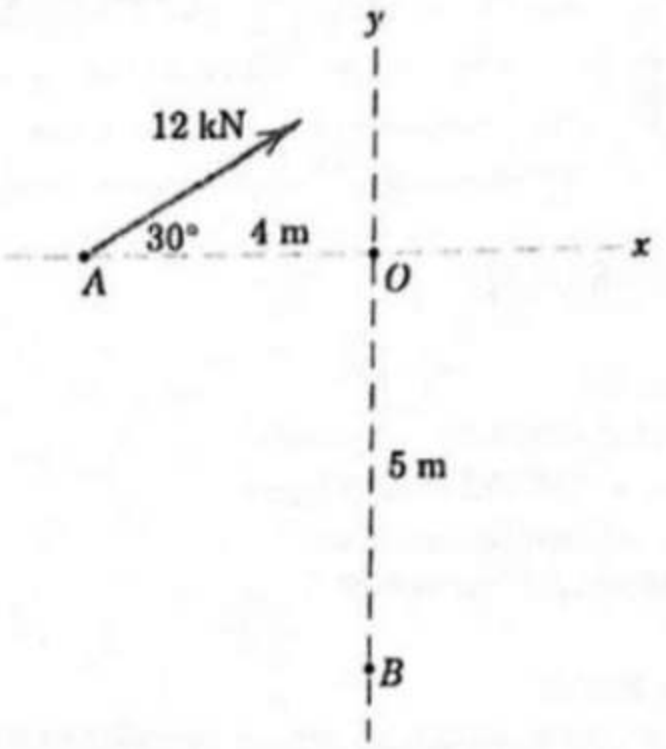
Subject Code: BMA 110	**Subject: Engineering Mechanics**
Time :1 ½Hours	**Maximum Marks : 30**
Note:Q. 1 is compulsory.	

Q1		**(5*2)**	
	(A) Select one answer choice i) Wedges are used to: a) Reduce friction between two surfaces b) Increase the force applied to an object c) Divide a force into two components d) Increase the stability of an object ii)The angle of repose is: a) The maximum angle at which an object remains at rest on an inclined plane b) The angle at which an object starts sliding on an inclined plane c) The angle at which the coefficient of friction is maximum d) The angle at which the force of friction is zero iii)The moment of inertia of a body is always: a) Positive b) Negative c) Zero d) Either positive or zero iv)The parallel axis theorem states that the moment of inertia of a body about an axis parallel to the axis through the centroid is: a) Greater than the moment of inertia about the centroidal axis b) Smaller than the moment of inertia about the centroidal axis c) Equal to the moment of inertia about the centroidal axis d) Not related to the moment of inertia about the centroidal axis v)The centroid of a composite body can be found by: a) Taking the average of all the centroids of its individual components b) Taking the sum of all the centroids of its individual components c) Taking the weighted average of all the centroids of its individual components d) Taking the centroid of its most significant component		
	(B)True or False: i) A particle is a body with a fixed shape and size. ii) Changing the orientation of a beam can cause a change in its resistance to bending deflections. iii) Changing the material of a beam can cause a change in its resistance to bending deflections. iv) Principle of Transmissibility is applicable to rigid bodies ONLY. v) Race car tires are made bigger in size to increase coefficient of friction against the road.		

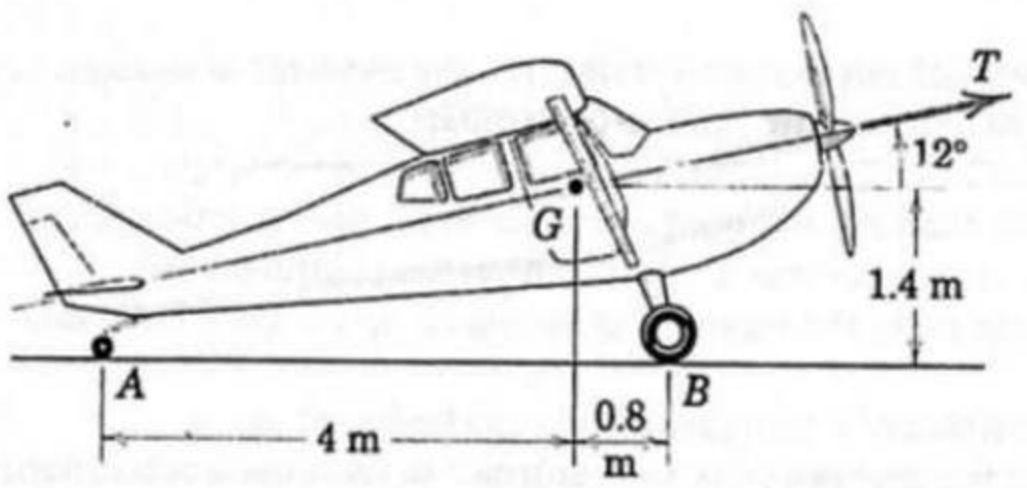
PTO

Q2	(Attempt any Two Parts) UNIT-1	(5,5)

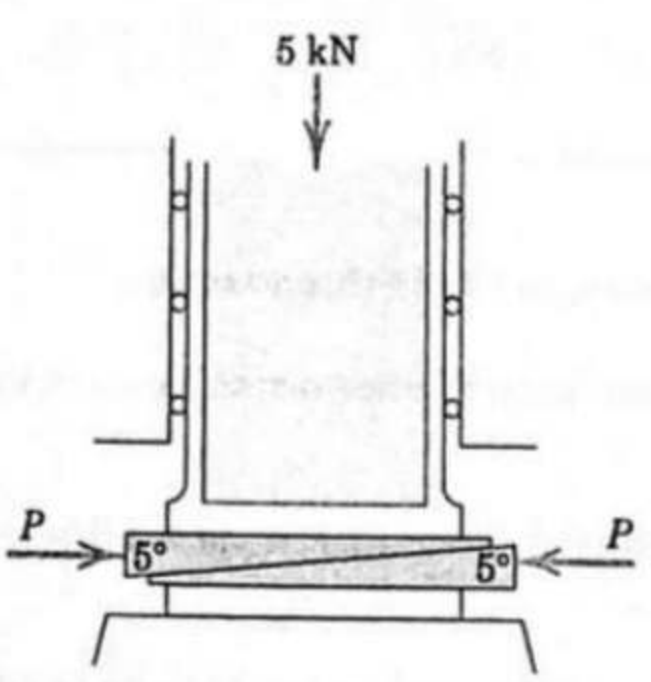
(a) Replace the 12-kN force acting at point A by a force–couple system at
(i) point O and
(ii) point B.



(b) During an engine test on the ground, a propeller thrust T = 3000 N is generated on the 1800-kg air-plane with mass center at G. The main wheels at B are locked and do not skid; the small tail wheel at A has no brake. Compute the percent change 'n' in the normal forces at A and B as compared with their "engine-off" values.

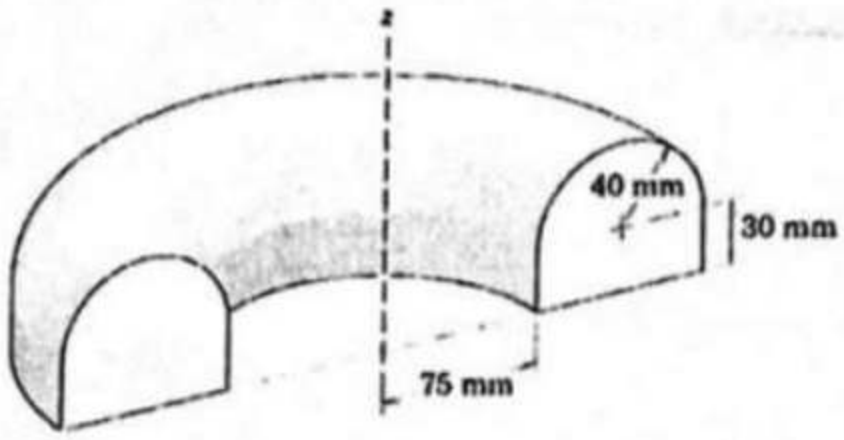


(c) The two 5 degrees wedges shown are used to adjust the position of the column under a vertical load of 5 kN. Determine the magnitude of the forces P required to raise the column if the coefficient of friction for all surfaces is 0.40.

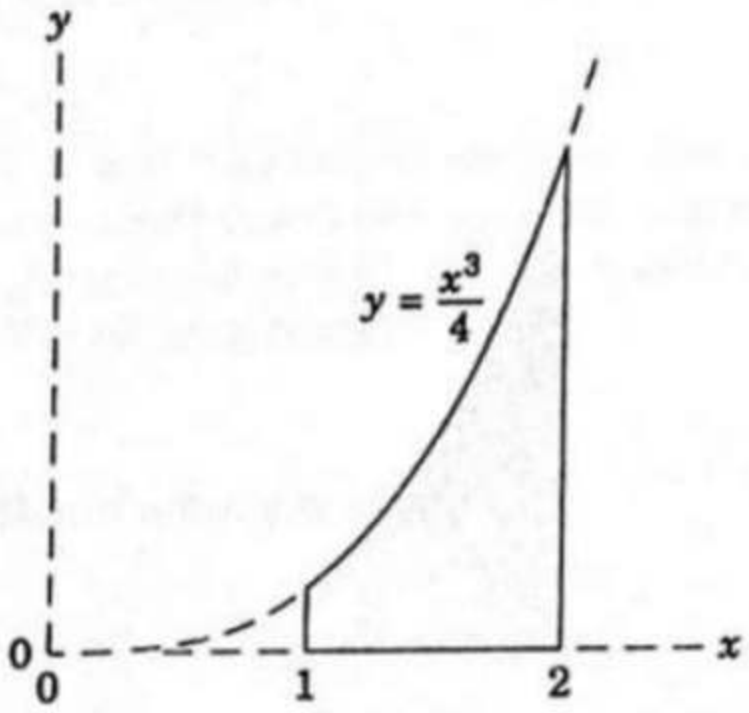


Q3	(Attempt any Two Parts) UNIT-2	(5,5)	

(a) Determine the volume V and total surface area A of the solid generated by revolving the area shown through 180 degrees about the z-axis, USING PAPPUS'S THEOREM.



(b) Determine the rectangular and polar radii of gyration of the shaded area about the axes shown.



(c) The mass per unit length of the slender rod varies with position according to $P=P_0(1-x^2/2)$ where x is in feet. Determine the location of the center of mass of the rod.

